Muhannad Yazbak

Full-Stack & Software Engineer | Dual B.Sc. (Information Systems & Medical Lab Sciences)

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PROFESSIONAL SUMMARY

Versatile Software Engineer with a dual-degree background in Information Systems and Medical Laboratory Sciences. Expert in building scalable full-stack web applications, multi-tenant SaaS platforms, microservices, and AI-driven solutions using React, Next.js, Node.js, Python, and PostgreSQL. Combines 2.5 years of clinical diagnostic rigor with robust software architecture, RESTful API design, and automated testing to deliver resilient, production-ready systems.

TECHNICAL SKILLS

- **Languages & Core**: TypeScript, JavaScript, Python, Java, SQL, REST APIs
- **Frontend & Desktop**: React.js, Next.js, Electron, Vite, Tailwind CSS
- **Backend & AI**: Node.js, Express.js, FastAPI, RAG Microservices, LangChain, OpenRouter API
- **Databases & ORM**: PostgreSQL, MySQL, SQLite, SQLAlchemy, MongoDB, Mongoose
- **DevOps & Cloud**: Docker, Kubernetes (kind), GitHub Actions CI/CD, Vercel, Render, Aiven
- **QA & Testing**: Playwright (POM), Vitest, E2E Testing, Allure Reporting

COMMERCIAL & FREELANCE EXPERIENCE

Freelance Full-Stack & Software Engineer | Self-Employed | 2023 - Present

• Multi-Tenant Vertical CRM Platform: Engineered a multi-tenant SaaS platform featuring vertical-specific workflows, RBAC, dynamic PostgreSQL database schemas, and automated Playwright test pipelines in GitHub Actions.

• E-Commerce Web Ecosystem: Built a decoupled, production web platform separating a Next.js/Vercel frontend from a containerized FastAPI backend, featuring an asynchronous Python RAG microservice for semantic product search.

• Localized Agency ERP (Desktop): Architected a local-first Electron & SQLite desktop CRM/ERP for a local business to manage client records, policy lifecycles, and document ingestion via Node.js file handlers.

TECHNICAL PROJECTS

Autonomous Playwright Self-Healing Test Agent | Python, Playwright, LLM APIs, GitHub Actions | 2026

• Developed an AI-driven test system that intercepts broken DOM locators at runtime and passes context to an LLM for real-time selector repair, drastically reducing test maintenance overhead.

• Automated exploratory session tracking to auto-generate native spec.ts test files and optimized CI/CD caching.

EDUCATION & PROFESSIONAL TRAINING

• 2019 - 2023: B.Sc. in Information Systems (BI Specialization) — University of Haifa

• 2021 - 2022: Software Engineering Training — Tsofen (Java, Spring Boot) & WebAhead (React, JS)

• 2023: QA Automation Engineering Program — Appleseeds (Full E2E suite)

• 2010 - 2013: B.Sc. in Medical Laboratory Sciences — Ben-Gurion University of the Negev (GPA: 88)

PREVIOUS PROFESSIONAL EXPERIENCE

Medical Laboratory Scientist / Specialist | Clinical Laboratories | 2016 - 2019

• Executed high-throughput diagnostic analyses, maintained strict protocol compliance, and managed complex medical data validation routines—establishing the diagnostic accuracy applied to backend system architecture.